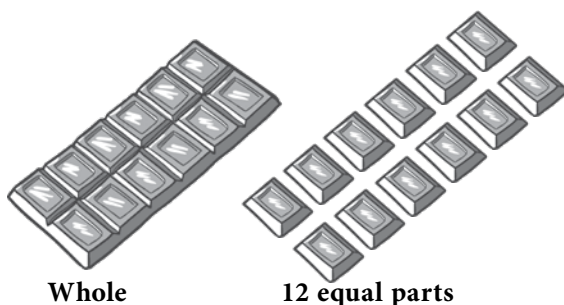


Learning objectives**3.1** Fraction**3.3** Fraction of shaded and unshaded parts**3.5** Comparison of fractions**3.2** Fraction of a collection**3.4** Types of fractions**3.6** Addition and subtraction of like fractions**3.1 FRACTION**

When a thing is divided into equal parts, then each part is called a fraction of the whole.

A fraction is written in the form as $\frac{\text{Numerator}}{\text{Denominator}}$.

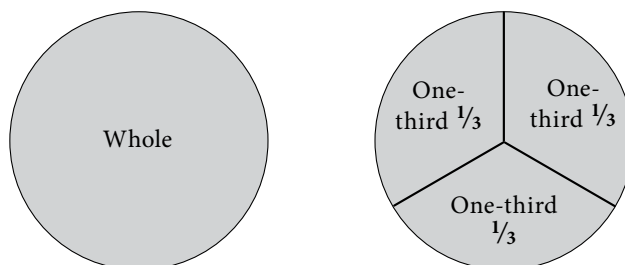
Here, numerator represents equal parts being talked about and denominator is the total number of equal parts.

**One-Half**

When a whole is divided into two equal parts, then each part is called the one-half $\left(\frac{1}{2}\right)$ of the whole.

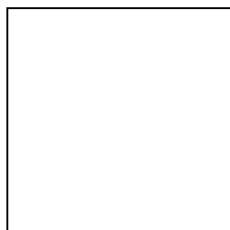
**One-Third**

When a whole is divided into three equal parts, then each part is called the one-third $\left(\frac{1}{3}\right)$ of the whole.

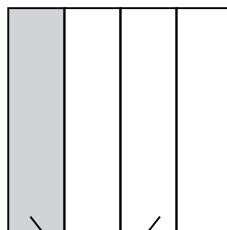


One-Fourth

When a whole is divided into four equal parts, then each part is called one-fourth $\left(\frac{1}{4}\right)$ of the whole.



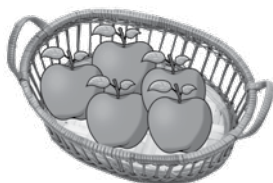
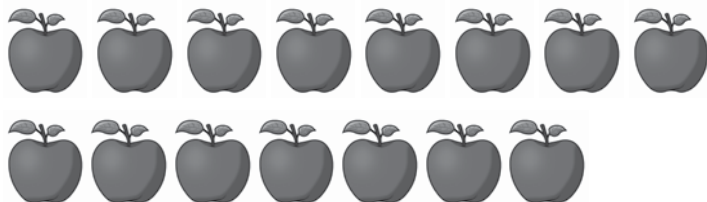
Whole



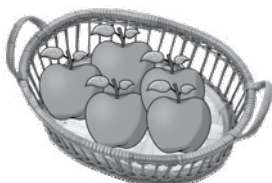
One-fourth

3.2 FRACTION OF A COLLECTION

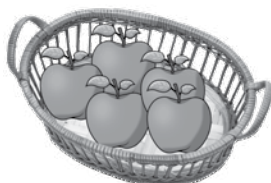
Aman had a collection of 15 apples. He want to put these apples in three basket in such a way that there are equal number of apples in all the baskets.



5 Apples



5 Apples



5 Apples

So, $\frac{1}{3}$ of 15 = $15 \div 3 = 5$

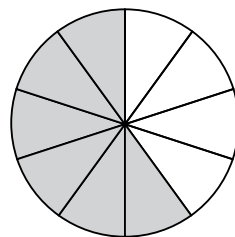
3.3 FRACTION OF SHADED AND UNSHADED PARTS

$$\text{Fraction of shaded/unshaded part} = \frac{\text{Number of shaded/unshaded parts}}{\text{Total number of equal parts}}$$

Consider the given figure.

$$\text{Fraction for shaded part} = \frac{\text{Number of shaded parts}}{\text{Total number of equal parts}} = \frac{6}{10}$$

$$\text{Fraction for unshaded part} = \frac{\text{Number of unshaded parts}}{\text{Total number of equal parts}} = \frac{4}{10}$$

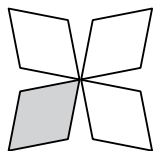


Olympiad Bite

- To calculate the fractional part of any quantity we divide objects into groups equal to the denominator.
- $\frac{1}{2}$ of a collection $\rightarrow$ Divide whole by 2.
- $\frac{1}{3}$ of a collection $\rightarrow$ Divide whole by 3.
- $\frac{1}{4}$ of a collection $\rightarrow$ Divide whole by 4.

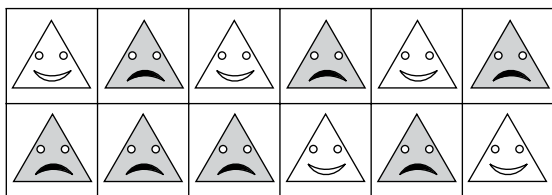
SELF TEST - 1

1. Find the shaded fraction of the given figure.



- (A) $\frac{2}{4}$ (B) $\frac{1}{4}$
(C) $\frac{3}{4}$ (D) 1

2. What fraction of the faces are smiling?



- (A) $\frac{5}{8}$ (B) $\frac{1}{2}$

(C) $\frac{5}{12}$

(D) $\frac{7}{12}$

3. Denominator of the fraction $\frac{4}{11}$ is _____.

- (A) 4 (B) 6
(C) 11 (D) 5

4. One-sixth of a collection of 24 toys is _____.

- (A) 12 (B) 4
(C) 18 (D) 16

5. Find the value of the following.

$$\frac{1}{3} \text{ of } 36$$

- (A) 18 (B) 15
(C) 12 (D) 9

3.4 TYPES OF FRACTIONS

Unit Fraction

A fraction in which the numerator is 1 is called unit fraction.

For example : $\frac{1}{2}, \frac{1}{5}, \frac{1}{7}$

Proper Fraction

A fraction in which numerator is smaller than the denominator is called proper fraction.

For example : $\frac{3}{4}, \frac{5}{8}, \frac{7}{9}$

Improper Fraction

A fraction in which numerator is greater than the denominator is called improper fraction.

For example : $\frac{6}{5}, \frac{9}{7}, \frac{5}{3}$

Like Fractions

Fractions having same denominator but different numerators.

For example : $\frac{4}{7}$ and $\frac{6}{7}$ are like fractions.



Olympiad Bite

- All unit fractions are proper fractions.
- There is no fraction with 0 as denominator.

Unlike Fractions

Fractions having different denominators. (It doesn't matter what the numerator is)

For example : $\frac{5}{7}$ and $\frac{9}{4}$ are unlike fractions.

3.5 COMPARISON OF FRACTIONS

- When the numerators are same, then the number with smaller denominator is greater.

For example : $\frac{2}{4} > \frac{2}{6} > \frac{2}{7} > \frac{2}{8}$

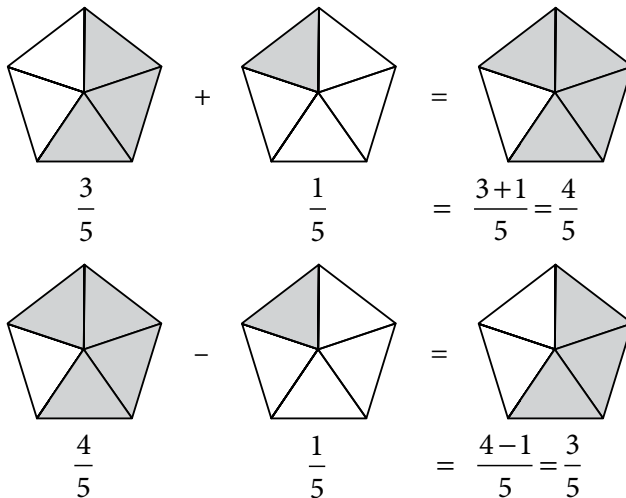
- When the denominators are same, then the number with greater numerator is greater.

For example : $\frac{7}{3} > \frac{6}{3} > \frac{5}{3} > \frac{4}{3}$

3.6 ADDITION AND SUBTRACTION OF LIKE FRACTIONS

In addition/subtraction of like fractions only numerators are added/subtracted.

For example :



SELF TEST - 2

1. Which of the following is a unit fraction?

(A) $\frac{4}{5}$ (B) $\frac{2}{3}$
(C) $\frac{3}{2}$ (D) $\frac{1}{5}$

2. Find the sum of $\frac{2}{5}$ and $\frac{6}{5}$.

(A) $\frac{4}{5}$ (B) $\frac{8}{5}$
(C) $\frac{3}{5}$ (D) $\frac{1}{5}$

3. Compare and fill in the box.

$$\frac{4}{7} + \frac{2}{7} \square \frac{5}{7} - \frac{2}{7}$$

(A) <

(B) >

(C) =

(D) Can't be determined

4. Which of the following fractions is greatest?

(A) $\frac{3}{11}$ (B) $\frac{4}{11}$

(C) $\frac{8}{11}$ (D) $\frac{9}{11}$

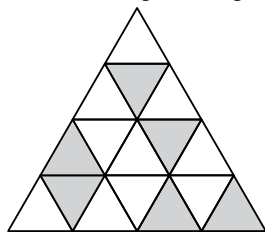
5. Which of the following set of fractions is like fractions?

(A) $\frac{2}{3}, \frac{4}{5}, \frac{6}{7}, \frac{7}{8}$ (B) $\frac{4}{5}, \frac{3}{5}, \frac{2}{5}, \frac{1}{5}$

(C) $\frac{5}{6}, \frac{5}{7}, \frac{5}{8}, \frac{5}{9}$ (D) $\frac{1}{2}, \frac{3}{4}, \frac{4}{5}, \frac{5}{7}$

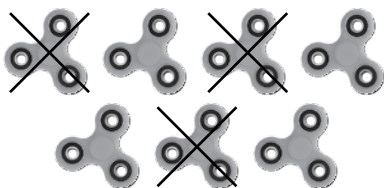
EXERCISE

1. What fraction of the given figure is shaded?



- (A) $\frac{8}{16}$ (B) $\frac{6}{16}$
(C) $\frac{10}{16}$ (D) $\frac{5}{16}$

2. Which of the following shows the fraction of crossed fidget?



- (A) $\frac{7}{3}$ (B) $\frac{5}{7}$
(C) $\frac{4}{7}$ (D) $\frac{3}{7}$

3. Which of the following options shows the correct descending order of fractions?

- (A) $\frac{4}{7}, \frac{5}{7}, \frac{8}{7}, \frac{6}{7}$ (B) $\frac{3}{10}, \frac{4}{10}, \frac{5}{10}, \frac{7}{10}$
(C) $\frac{8}{9}, \frac{7}{9}, \frac{5}{9}, \frac{2}{9}$ (D) None of these

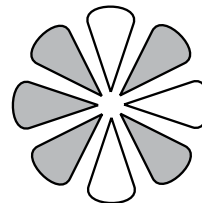
4. Which of the following is a proper fraction?

- (A) $\frac{17}{9}$ (B) $\frac{3}{5}$
(C) $\frac{12}{7}$ (D) $\frac{11}{3}$

5. Subtract $\frac{4}{13}$ from $\frac{9}{13}$.

- (A) $\frac{5}{13}$ (B) $\frac{3}{13}$
(C) $\frac{4}{13}$ (D) $\frac{8}{13}$

6. Fraction of the shaded parts of the given figure is ____.

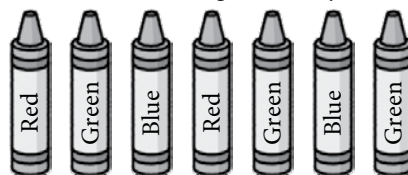


- (A) $\frac{1}{8}$ (B) $\frac{7}{8}$
(C) $\frac{5}{8}$ (D) $\frac{3}{8}$

7. A pizza was cut into 12 equal pieces. Aanya ate 5 pieces. What fraction of the pizza is left with her?

- (A) $\frac{5}{12}$ (B) $\frac{7}{12}$
(C) $\frac{3}{12}$ (D) $\frac{12}{5}$

8. What fraction of the given crayons are green?



- (A) $\frac{1}{7}$ (B) $\frac{2}{7}$
(C) $\frac{4}{7}$ (D) $\frac{3}{7}$

9. $\frac{3}{4}$ represent ____.

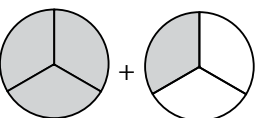
- (A) A whole (B) 3 parts of whole
(C) 4 parts of whole (D) 7 parts of whole

10. Aanchal's birthday cake was cut into 10 equal pieces. She distribute them equally among her five friends. What fraction of the cake each friend get?

- (A) $\frac{1}{10}$ (B) $\frac{5}{10}$
(C) $\frac{2}{10}$ (D) $\frac{8}{10}$

11. Rashmi bakes 37 biscuits. She gave 23 biscuits to poor children. What fraction of biscuits is left with her?

- (A) $\frac{14}{37}$ (B) $\frac{23}{37}$
(C) $\frac{15}{37}$ (D) $\frac{37}{14}$

12. Shaded fraction of  = ?

- (A) $\frac{5}{3}$ (B) $\frac{7}{3}$
(C) $\frac{4}{3}$ (D) $\frac{8}{3}$

13. There are 8 , 5 , 3  in a zoo. What fraction of the total animals are monkey in the zoo?

- (A) $\frac{3}{15}$ (B) $\frac{5}{16}$
(C) $\frac{3}{16}$ (D) $\frac{8}{15}$

14. What fraction of letters in the word GEOMETRY are vowel?

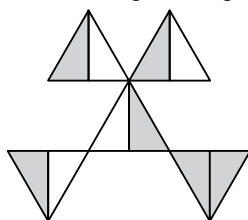
- (A) $\frac{2}{8}$ (B) $\frac{3}{8}$
(C) $\frac{2}{7}$ (D) $\frac{4}{8}$

15. Fraction of the digit 7 in the given series is _____.

2 7 9 8 1 5 4 7 3 2 5 7 8 9 1 0 5 4 7 9 8

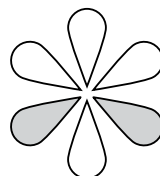
- (A) $\frac{5}{21}$ (B) $\frac{3}{19}$
(C) $\frac{4}{17}$ (D) $\frac{4}{21}$

16. What fraction of the given figure is unshaded?



- (A) $\frac{5}{10}$ (B) $\frac{4}{10}$
(C) $\frac{6}{10}$ (D) $\frac{7}{10}$

17. The flower shown contains 6 petals. How many more petals must be shaded so that $\frac{1}{2}$ of the flower is shaded?

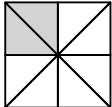
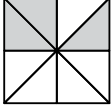
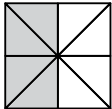
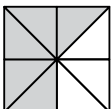


- (A) 1 (B) 2
(C) 3 (D) 4

18. There is no fraction whose denominator is _____.

- (A) 0 (B) 1
(C) 2 (D) 5

19. Which of the following options shows the correct match of shaded fraction?

- (A)  = $\frac{5}{8}$ (B)  = $\frac{4}{5}$
(C)  = $\frac{4}{8}$ (D)  = $\frac{3}{8}$

20. Which symbol will make the given number sentence true?

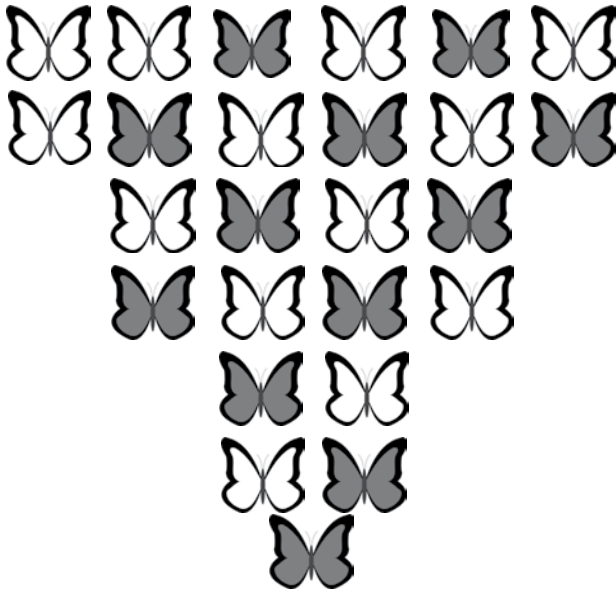
$$\frac{2}{7} + \frac{4}{7} - \frac{1}{7} \square \frac{1}{7} + \frac{5}{7} - \frac{6}{7}$$

- (A) > (B) <
(C) = (D) Can't be determined

21. Sneha puts $\frac{1}{5}$ cup of sugar in a bowl. Then she adds $\frac{3}{5}$ cup of sugar more. How much sugar is in the bowl now?

- (A) $\frac{3}{5}$ cup (B) $\frac{4}{5}$ cup
(C) $\frac{2}{5}$ cup (D) 1 cup

22. What fraction of the given butterflies are not shaded?



- (A) $\frac{12}{25}$ (B) $\frac{13}{25}$
(C) $\frac{21}{26}$ (D) $\frac{5}{25}$

23. What fraction of English alphabets is consonant?

- (A) $\frac{5}{26}$ (B) $\frac{5}{21}$
(C) $\frac{21}{26}$ (D) $\frac{5}{24}$

24. There are 60 trees in a garden. Kirti has watered $\frac{1}{3}$ of the trees. What fraction of trees unwatered?

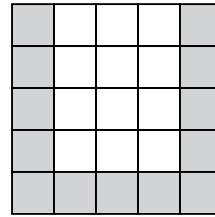
- (A) $\frac{40}{60}$ (B) $\frac{5}{60}$
(C) $\frac{21}{26}$ (D) $\frac{1}{60}$

25. What fraction of stars are outside the box?



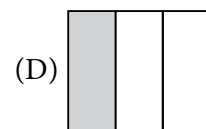
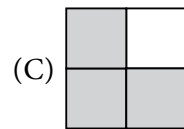
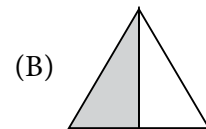
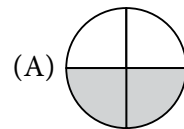
- (A) $\frac{5}{13}$ (B) $\frac{5}{8}$
(C) $\frac{8}{13}$ (D) $\frac{8}{5}$

26. The fraction of unshaded region shown in the figure is represented by _____.



- (A) $\frac{12}{13}$ (B) $\frac{13}{25}$
(C) $\frac{13}{12}$ (D) $\frac{12}{25}$

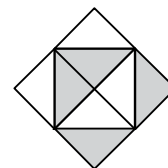
27. Which of the following shapes shows $\frac{1}{3}$ part shaded?



28. Piyush bought 85 cans of paint and used 32 cans. What fraction of the paint of cans did he use?

- (A) $\frac{32}{85}$ (B) $\frac{35}{85}$
(C) $\frac{53}{85}$ (D) $\frac{32}{53}$

29. What fraction of the given figure is shaded?



- (A) $\frac{4}{8}$ (B) $\frac{4}{7}$
(C) $\frac{4}{6}$ (D) $\frac{3}{8}$

30. Vaibhav has fifty ₹ 10 coins and twenty five ₹ 5 coins. What fraction of coins are ₹ 10 coins?

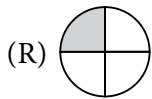
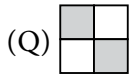
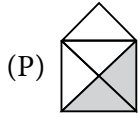
- (A) $\frac{25}{75}$ (B) $\frac{25}{50}$
(C) $\frac{50}{75}$ (D) $\frac{50}{100}$

Achievers Section (HOTS)

31. Match the figures given in Column 1 with their shaded fractions given in Column 2.

Column 1

Column 2



(1) $\frac{1}{4}$

(2) $\frac{2}{5}$

(3) $\frac{2}{4}$

- (A) (P) → (1); (Q) → (3); (R) → (2)
 (B) (P) → (2); (Q) → (3); (R) → (1)
 (C) (P) → (2); (Q) → (1); (R) → (3)
 (D) (P) → (1); (Q) → (2); (R) → (3)

32. Fill in the blanks.

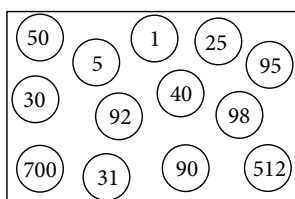
- (P) The figure represents _____ shaded fraction.

- (Q) In the figure _____ fraction of the bananas are encircled.

- (R) _____ fraction of letters in the word CREATIVE are vowels.

- | (P) | (Q) | (R) |
|-------------------|---------------|---------------|
| (A) $\frac{4}{8}$ | $\frac{4}{8}$ | $\frac{3}{8}$ |
| (B) $\frac{5}{8}$ | $\frac{4}{8}$ | $\frac{4}{8}$ |
| (C) $\frac{5}{8}$ | $\frac{4}{8}$ | $\frac{3}{8}$ |
| (D) $\frac{4}{8}$ | $\frac{5}{8}$ | $\frac{4}{8}$ |

33. Which of the following statements is CORRECT about the given figure?



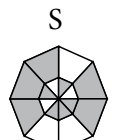
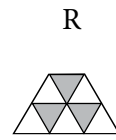
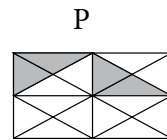
- (A) Fraction of numbers having zero at units place is $\frac{5}{13}$.

- (B) Fraction of numbers having nine at tens place is $\frac{4}{13}$.

- (C) Fraction of three digit numbers is $\frac{2}{13}$.

- (D) All of these

34. Arrange the shaded fractions of given figures in descending order.



- (A) P, Q, R, S
 (B) S, Q, R, P
 (C) P, Q, S, R
 (D) S, R, P, Q

35. If = and = 1 whole, then = ?.

- (A) $\frac{2}{9}$
 (B) $\frac{4}{9}$
 (C) $\frac{8}{9}$
 (D) $\frac{1}{9}$

SOF IMO 2019 QUESTIONS

1. What fraction of the given butterflies is NOT red?



- (A) $\frac{6}{11}$
 (B) $\frac{5}{11}$
 (C) $\frac{4}{11}$
 (D) $\frac{2}{11}$ (Level-1)

2. Malika has 10 strawberry ice-creams and 15 chocolate ice-creams. What fraction of the total ice-creams is the chocolate ice-creams?



Strawberry



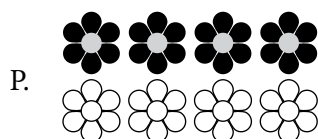
Chocolate

- (A) $\frac{5}{25}$ (B) $\frac{15}{25}$
 (C) $\frac{10}{15}$ (D) $\frac{10}{25}$ (Level-1)

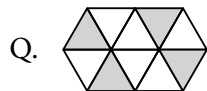
3. Match the following figures with their correct shaded fraction and select the correct option.

Column A

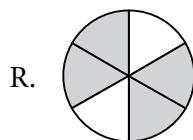
Column B



(a) $\frac{4}{10}$



(b) $\frac{4}{6}$



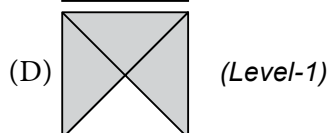
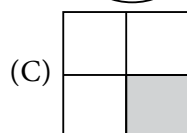
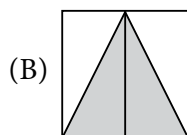
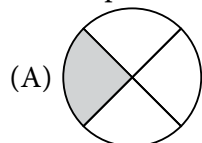
(c) $\frac{4}{8}$

- (A) P → (c), Q → (b), R → (a)
 (B) P → (b), Q → (c), R → (a)
 (C) P → (c), Q → (a), R → (b)
 (D) P → (b), Q → (a), R → (c) (Level-1)

4. In which of the following options, fractions are arranged in ascending order?

- (A) $\frac{2}{7}, \frac{9}{7}, \frac{5}{7}, \frac{3}{7}$ (B) $\frac{2}{7}, \frac{9}{7}, \frac{3}{7}, \frac{5}{7}$
 (C) $\frac{2}{7}, \frac{3}{7}, \frac{5}{7}, \frac{9}{7}$ (D) $\frac{9}{7}, \frac{5}{7}, \frac{3}{7}, \frac{2}{7}$ (Level-1)

5. Which of the following figures shows $\frac{2}{4}$ shaded part?



6. There were 1010 pastries in a shop. Of these, 175 were butterscotch pastries, 240 were chocolate and the rest were pineapple. What fraction of pastries

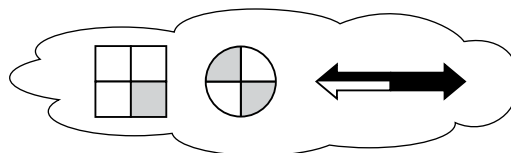
were pineapple in the shop?

- (A) $\frac{275}{1010}$ (B) $\frac{415}{1010}$
 (C) $\frac{119}{1010}$ (D) $\frac{595}{1010}$ (Level-1)

7. What fraction of the letters of the word LIBRARY is vowel?

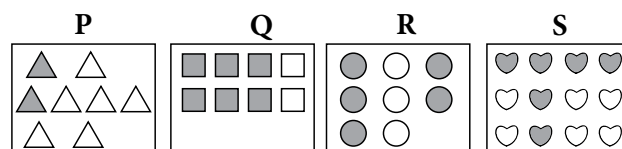
- (A) $\frac{5}{7}$ (B) $\frac{7}{2}$
 (C) $\frac{2}{7}$ (D) $\frac{7}{5}$ (Level-2)

8. Find the sum of shaded fractions of the given figures.



- (A) $\frac{8}{4}$ (B) $\frac{5}{4}$
 (C) $\frac{7}{4}$ (D) $\frac{6}{4}$ (Level-2)

9. Which of the following has $\frac{6}{8}$ of its contents shaded?



- (A) P (B) Q
 (C) R (D) S (Level-2)

10. Match the following and select the correct option.

Column-A

Column-B

- P. $\frac{7}{9} - \frac{6}{9}$ (a) $\frac{9}{13}$
 Q. $\frac{1}{4} + \frac{3}{4} + \frac{2}{4}$ (b) $\frac{1}{9}$
 R. $\frac{3}{13} + \frac{7}{13} - \frac{1}{13}$ (c) $\frac{2}{14}$
 S. $\frac{12}{14} - \frac{10}{14}$ (d) $\frac{6}{4}$

- (A) P-(a); Q-(d); R-(c); S-(b)
 (B) P-(b); Q-(d); R-(a); S-(c)
 (C) P-(b); Q-(a); R-(d); S-(c)
 (D) P-(c); Q-(d); R-(a); S-(b) (Level-2)

HINTS & EXPLANATIONS

SELF TEST - 1

1. (B): Total number of equal parts = 4

Number of shaded part = 1

So, shaded fraction = $\frac{1}{4}$

2. (C): Total number of faces = 12

Number of smiling faces = 5

∴ Required fraction = $\frac{5}{12}$

3. (C): Fraction = $\frac{4}{11}$ → Numerator
→ Denominator

4. (B): Total number of toys = 24

So, one-sixth of 24 toys = $24 \div 6 = 4$

5. (C): $\frac{1}{3}$ of 36 = $36 \div 3 = 12$

SELF TEST - 2

1. (D)

2. (B): Required sum = $\frac{2}{5} + \frac{6}{5} = \frac{2+6}{5} = \frac{8}{5}$

3. (B): $\frac{4}{7} + \frac{2}{7} = \frac{4+2}{7} = \frac{6}{7}$

$\frac{5}{7} - \frac{2}{7} = \frac{5-2}{7} = \frac{3}{7}$

And $\frac{6}{7} > \frac{3}{7}$

4. (D): $\frac{9}{11} > \frac{8}{11} > \frac{4}{11} > \frac{3}{11}$

5. (B)

EXERCISE

1. (B): Total number of equal parts = 16

Number of shaded parts = 6

So, required shaded fraction = $\frac{6}{16}$

2. (D): Total number of fidgets = 7

Number of crossed fidgets = 3

So, required fraction = $\frac{3}{7}$

3. (C): $\frac{8}{9} > \frac{7}{9} > \frac{5}{9} > \frac{2}{9}$

4. (B)

5. (A): Required difference = $\frac{9}{13} - \frac{4}{13} = \frac{9-4}{13} = \frac{5}{13}$

6. (C): Total number of equal parts = 8

Number of shaded parts = 5

So, shaded fraction = $\frac{5}{8}$

7. (B): Total number of equal parts of pizza = 12

Number of parts of pizza Aanya ate = 5

Number of parts of pizza left = $12 - 5 = 7$

So, fraction of pizza left = $\frac{7}{12}$

8. (D): Total number of crayons = 7

Number of green crayons = 3

So, fraction of green crayons = $\frac{3}{7}$

9. (B)

10. (C): Total number of equal pieces of cake = 10

Number of friends = 5

So, each friend get $10 \div 5 = 2$ equal pieces

So, fraction of cake each friend get = $\frac{2}{10}$

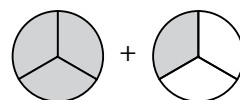
11. (A): Total number of biscuits baked = 37

Number of biscuits given to poor children = 23

Number of biscuits left = $37 - 23 = 14$

So, fraction of biscuits left = $\frac{14}{37}$

12. (C): Shaded fraction of



+ $\frac{3}{3} + \frac{1}{3} = \frac{3+1}{3} = \frac{4}{3}$

13. (C): Total number of animals in the zoo

= $8 + 5 + 3 = 16$

Number of monkeys in the zoo = 3

So, required fraction = $\frac{3}{16}$

14. (B): Total number of letters in the given word

= 8

Total number of vowels (E, O, E) = 3

So, fraction of vowels in the given word = $\frac{3}{8}$

15. (D): Total number of digits in the given series = 21

Number of digit 7 in the given series = 4

So, required fraction = $\frac{4}{21}$

16. (B): Total number of equal parts = 10

Number of unshaded parts = 4

So, fraction of unshaded part = $\frac{4}{10}$

17. (A): Total number of petals = 6

Total number of petals must be shaded to make

shaded fraction $\frac{1}{2} = \frac{1}{2} \times 6 = 3$

Number of more petals must be shaded = $3 - 2 = 1$

18. (A)

19. (C):

	Number of equal parts	Number of shaded parts	Shaded fraction
(A)	8	2	2/8
(B)	8	3	3/8
(C)	8	4	4/8
(D)	8	5	5/8

20. (A): $\frac{2}{7} + \frac{4}{7} - \frac{1}{7} = \frac{2+4}{7} - \frac{1}{7} = \frac{6}{7} - \frac{1}{7} = \frac{6-1}{7} = \frac{5}{7}$

and $\frac{1}{7} + \frac{5}{7} - \frac{6}{7} = \frac{1+5}{7} - \frac{6}{7} = \frac{6}{7} - \frac{6}{7} = \frac{6-6}{7} = \frac{0}{7} = 0$

So, $\frac{5}{7} > 0$

21. (B): Quantity of sugar Sneha puts in bowl = $\frac{1}{5}$ cup

Quantity of sugar added = $\frac{3}{5}$ cup

Quantity of sugar in bowl now = $\left(\frac{1}{5} + \frac{3}{5}\right)$ cup
 $= \frac{1+3}{5}$ cup = $\frac{4}{5}$ cup

22. (B): Total number of butterflies = 25

Number of shaded butterflies = 12

So, number of butterflies not shaded = $25 - 12 = 13$

So, fraction of butterflies not shaded = $\frac{13}{25}$

23. (C): Total number of alphabets = 26

Total number of consonants = 21

So, fraction of consonants = $\frac{21}{26}$

24. (A): Total number of trees in the garden = 60

Number of trees Kirti watered = $\frac{1}{3}$ of 60

= $60 \div 3 = 20$

$\therefore$ Number of trees unwatered = $60 - 20 = 40$

So, fraction of trees left unwatered = $\frac{40}{60}$

25. (A): Total number of stars = 13

Number of stars outside the box = 5

So, required fraction = $\frac{5}{13}$

26. (D): Total number of equal parts = 25

Number of unshaded parts = 12

So, fraction of unshaded region = $\frac{12}{25}$

27. (D): (A) Total number of equal parts = 4

Number of shaded parts = 2

$\therefore$ Shaded fraction = $\frac{2}{4}$

(B) Total number of equal parts = 2

Number of shaded part = 1

$\therefore$ Shaded fraction = $\frac{1}{2}$

(C) Total number of equal parts = 4

Number of shaded parts = 3

$\therefore$ Shaded fraction = $\frac{3}{4}$

(D) Total number of equal parts = 3

Number of shaded part = 1

$\therefore$ Shaded fraction = $\frac{1}{3}$

28. (A): Total number of cans bought = 85

Number of cans used = 32

So, fraction of cans used = $\frac{32}{85}$

29. (A): Total number of equal parts = 8

Number of shaded parts = 4

So, fraction of shaded part = $\frac{4}{8}$

30. (C): Total number of coins = $50 + 25 = 75$

Number of ₹ 10 coins = 50

So, fraction of ₹ 10 coins = $\frac{50}{75}$

31. (B): (P) Total number of equal parts = 5

Number of shaded parts = 2

So, shaded fraction = $\frac{2}{5}$

SOF IMO 2019 QUESTIONS

(Q) Total number of equal parts = 4

Number of shaded parts = 2

So, shaded fraction = $\frac{2}{4}$

(R) Total number of equal parts = 4

Number of shaded part = 1

So, shaded fraction = $\frac{1}{4}$

32. (B): (P) Total number of equal parts = 8

Number of shaded parts = 5

So, shaded fraction = $\frac{5}{8}$

(Q) Total number of bananas = 8

Number of encircled bananas = 4

So, fraction of encircled bananas = $\frac{4}{8}$

(R) Total number of letters = 8

Number of vowels (E, A, I, E) = 4

So, fraction of vowels = $\frac{4}{8}$

33. (D): Total numbers = 13

(A) Numbers having zero at units place

(50, 40, 30, 90, 700) = 5

So, required fraction = $\frac{5}{13}$

(B) Numbers having 9 at tens place (92, 98, 90, 95) = 4

So, required fraction = $\frac{4}{13}$

(C) Three digit numbers (700, 512) = 2

So, required fraction = $\frac{2}{13}$

34. (B):

Figure	Shaded fraction
P	$\frac{2}{8}$
Q	$\frac{4}{8}$
R	$\frac{3}{8}$
S	$\frac{5}{8}$

Correct descending order is S, Q, R, P.

35. (B): 1  = 2  $\Rightarrow$ 3  = 6 

$\Rightarrow$ (6 + 3)  = 1 $\Rightarrow$ 9  = 1 $\Rightarrow$ 1  = $\frac{1}{9}$

$\Rightarrow$    =     = $\frac{4}{9}$

1. (A): Total number of butterflies = 11

Number of butterflies which are not red = 6

So, required fraction = $\frac{6}{11}$

2. (B): Total number of ice-creams = 10 + 15 = 25

Number of chocolate ice-creams = 15

So, required fraction = $\frac{15}{25}$

3. (C): (P) Total number of flowers = 8

Number of shaded flowers = 4

$\therefore$ Shaded fraction = $\frac{4}{8}$

(Q) Total number of equal parts = 10

Number of shaded parts = 4

$\therefore$ Shaded fraction = $\frac{4}{10}$

(R) Total number of equal parts = 6

Number of shaded parts = 4

$\therefore$ Shaded fraction = $\frac{4}{6}$

4. (C): $\frac{2}{7} < \frac{3}{7} < \frac{5}{7} < \frac{9}{7}$

5. (B): (A) Total number of equal parts = 4

Number of shaded parts = 1

So, shaded fraction = $\frac{1}{4}$

(B) Total number of equal parts = 4

Number of shaded parts = 2

So, shaded fraction = $\frac{2}{4}$

(C) Total number of equal parts = 4

Number of shaded parts = 1

So, shaded fraction = $\frac{1}{4}$

(D) Total number of equal parts = 4

Number of shaded parts = 3

So, shaded fraction = $\frac{3}{4}$

6. (D): Total number of pastries = 1010

Total number of butterscotch and chocolate pastries

= 175 + 240 = 415

$\therefore$ Number of pineapple pastries = 1010 - 415 = 595

So, fraction of pineapple pastries = $\frac{595}{1010}$

7. (C): Total number of letters = 7

Number of vowels (I, A) = 2

So, fraction of vowels = $\frac{2}{7}$

8. (D): Shaded fraction of first figure = $\frac{1}{4}$

Shaded fraction of second figure = $\frac{2}{4}$

Shaded fraction of third figure = $\frac{3}{4}$

∴ Required sum = $\frac{1}{4} + \frac{2}{4} + \frac{3}{4}$

$$= \frac{1+2+3}{4} = \frac{6}{4}$$

9. (B): P. Total number of figures = 8

Number of shaded figures = 2

So, shaded fraction = $\frac{2}{8}$

Q. Total number of figures = 8

Number of shaded figures = 6

So, shaded fraction = $\frac{6}{8}$

R. Total number of figures = 8

Number of shaded figures = 5

So, shaded fraction = $\frac{5}{8}$

S. Total number of figures = 12

Number of shaded figures = 6

So, shaded fraction = $\frac{6}{12}$

10. (B): (P) $\frac{7}{9} - \frac{6}{9} = \frac{7-6}{9} = \frac{1}{9}$

(Q) $\frac{1}{4} + \frac{3}{4} + \frac{2}{4} = \frac{1+3+2}{4} = \frac{6}{4}$

(R) $\frac{3}{13} + \frac{7}{13} - \frac{1}{13} = \frac{3+7}{13} - \frac{1}{13} = \frac{10}{13} - \frac{1}{13}$
 $= \frac{10-1}{13} = \frac{9}{13}$

(S) $\frac{12}{14} - \frac{10}{14} = \frac{12-10}{14} = \frac{2}{14}$